TATHAGATA ROY CHOWDHURY

Assistant Professor | PhD Researcher | Artificial Intelligence | Health Informatics | Explainable AI
Department of Computer Science and Engineering
Techno Engineering College, Banipur, West Bengal, India
PhD Researcher, National Institute of Technology Silchar, Assam, India
Email: tathagatamtech13@gmail.com | tathagata.r@tecb.edu.in
ORCID: 0000-0003-1052-4628 | Web of Science ResearcherID: QDN-0301-2026

Professional Summary

I am an Assistant Professor and PhD researcher in Computer Science and Engineering with more than a decade of experience in teaching, academic administration, research mentoring, curriculum development, technology management, and applied Artificial Intelligence. My doctoral research at the National Institute of Technology Silchar focuses on Health Informatics and Artificial Intelligence.

My research interests are centered on Artificial Intelligence, Machine Learning, Deep Learning, Medical Image Processing, Explainable Artificial Intelligence (XAI), Federated Learning, Healthcare Informatics, Data Security, Steganography, Generative AI, GANs, Diffusion Models, and AI-based clinical decision-support systems.

My current research activities include AI-based cancer prediction, lung-cancer prediction, PCOS prediction, privacy-preserving federated learning for genomic data, explainable healthcare AI, synthetic medical-image generation, and intelligent clinical systems.

Alongside research, I have extensive experience in academic leadership, departmental administration, ERP management, web administration, student project supervision, conference coordination, technical reviewing, session chairing, faculty development activities, and professional training.

Research Interests

- Artificial Intelligence and Machine Learning
- Deep Learning and Neural Networks
- Health Informatics and Healthcare AI
- Medical Image Processing and Analysis
- Explainable Artificial Intelligence (XAI)
- Federated Learning and Privacy-Preserving AI
- Generative AI, GANs and Diffusion Models
- Synthetic Medical Data Generation
- Strategic Synthetic Oversampling
- Cancer Prediction and Early Diagnosis
- Lung Cancer Prediction
- Breast Imaging and Histopathology Analysis
- PCOS Prediction and Clinical Analytics
- Colon Cancer Screening
- Clinical Decision Support Systems
- Natural Language Processing for Healthcare
- Data Security and Steganography
- Cryptography and Cloud Security
- Computer Architecture and Parallel Processing
- Intelligent Web and Healthcare Systems

Research Profile

My research profile combines theoretical Machine Learning and Deep Learning with practical healthcare applications. I am particularly interested in developing intelligent diagnostic frameworks that are accurate, explainable, privacy-preserving, and suitable for real-world healthcare environments.

My current research contributions include:

- **Advanced DeepLungCareNet:** A next-generation deep-learning framework for multiclass lung-cancer prediction.
- **DeepLungCareNet-FedWeb:** A federated-learning-enabled framework for privacy-preserving lung-

cancer diagnosis.

- **Bio-Cipher:** A privacy-preserving federated-learning architecture for genomic data analysis.
- **Intelligent PCOS Prediction Framework:** A CatBoost-based clinical prediction framework combined with automated clinical report generation.
- **Explainable Healthcare AI:** Research involving SHAP, LIME and Grad-CAM for transparent and interpretable healthcare prediction.
- **Synthetic Medical Image Generation:** Research involving Conditional GANs, Latent Diffusion Models and Strategic Synthetic Oversampling for medical-image classification.

Academic Identifiers and Research Profiles

Profile	Details
Google Scholar	Google Scholar Profile
ORCID	0000-0003-1052-4628
Web of Science	ResearcherID: QDN-0301-2026
Semantic Scholar	Tathagata Roy Chowdhury
ResearchGate	Tathagata Roy Chowdhury
Academia.edu	Tathagata Roy Chowdhury
LinkedIn	Tathagata Roy Chowdhury

Academic Qualifications

Qualification	Details
PhD (Pursuing)	Health Informatics, National Institute of Technology Silchar (NIT Silchar), Assam, India.
M.Tech in CSE	University of Calcutta, 2014.
B.Tech in CSE	Saroj Mohan Institute of Technology, 2011.

Professional Experience

Period	Position	Institution	Responsibilities
Mar 2024–Present	Assistant Professor	Techno Engineering College, Banipur	Teaching core Computer Science and Engineering subjects, academic leadership, student mentoring, research supervision, curriculum development, and AI/healthcare research.
Sept 2022–Mar 2024	Assistant Professor & In-Charge	St. Mary's Technical Campus, Kolkata	Departmental administration, academic coordination, faculty development, student activities, and institutional academic operations.
July 2022–Sept 2022	Assistant Professor	Brainware University	Teaching Machine Learning, programming, and Computer Science subjects with emphasis on practical and application-oriented pedagogy.

Period	Position	Institution	Responsibilities
Sept 2019–June 2022	Assistant Professor, Web Expert & ERP Manager	Elitte College of Engineering	Teaching, academic material development, institutional web management, and ORKA ERP administration.
Feb 2015–Sept 2019	Assistant Professor & Academic Advisor	Chaibasa Engineering College	Teaching, academic advising, technical seminars, workshops, student mentoring, and departmental coordination.

Academic Leadership and Institutional Contributions

- Served as Departmental In-Charge at St. Mary's Technical Campus, Kolkata.
- Served as Academic Advisor at Chaibasa Engineering College.
- Worked as Web Expert and ERP Manager at Elitte College of Engineering.
- Managed and supported institutional ORKA ERP systems.
- Contributed to institutional website development and administration.
- Coordinated faculty development and academic programmes.
- Developed academic materials and course resources.
- Mentored undergraduate students in technical projects and research.
- Guided students in research-paper preparation, publication, and presentation.
- Supported technical seminars, workshops, conferences and institutional events.
- Contributed to curriculum development and academic technology integration.

Technical Expertise

Area	Expertise
Artificial Intelligence	Machine Learning, Deep Learning, Explainable AI, Generative AI, predictive analytics.
Healthcare AI	Health Informatics, Medical Imaging, Clinical Decision Support, disease prediction and healthcare analytics.
Programming	Python, Java, C, C++, PHP.
Deep Learning Frameworks	TensorFlow, PyTorch.
Databases	Oracle Database and database-driven applications.
Web Technologies	Web application development, web administration and institutional web systems.
ERP	ORKA ERP management and institutional ERP implementation.
Research Technologies	Federated Learning, SHAP, LIME, Grad-CAM, GANs, Diffusion Models, NLP and medical-image analysis.

Current Research Projects and Frameworks

Advanced DeepLungCareNet

Advanced DeepLungCareNet is a next-generation deep-learning framework for multiclass lung-cancer prediction. The work focuses on developing intelligent diagnostic systems using advanced deep-learning methodologies for healthcare applications.

Published as a chapter in Springer Nature's Lecture Notes in Networks and Systems (LNNS) series in 2026.

DeepLungCareNet-FedWeb

DeepLungCareNet-FedWeb extends lung-cancer prediction into a federated-learning environment. The framework focuses on privacy-preserving collaborative healthcare intelligence while avoiding centralized sharing of sensitive medical information.

The work has been accepted at the Analytics Global Conference (AGC) 2026.

Bio-Cipher

Bio-Cipher is a privacy-preserving federated-learning architecture designed for genomic data analysis. The research focuses on secure distributed learning and privacy protection in sensitive biomedical datasets. The distributed loss was reduced from 0.56 to 0.51 in the reported experimental framework.

Intelligent PCOS Prediction Framework

The Intelligent PCOS Prediction Framework uses CatBoost-based machine learning for clinical prediction and incorporates automated clinical report generation.
DOI: 10.55041/ijsmt.v2i5.343.

Synthetic Over-Sampling for Medical Image Classification

I am developing a research monograph proposal titled:
Synthetic Over-Sampling for Medical Image Classification: From GANs and Diffusion Models to Clinical Deployment.
The research scope includes Conditional GANs, Latent Diffusion Models, Strategic Synthetic Oversampling (SSO), Grad-CAM, SHAP, Breast Ultrasound datasets, histopathology datasets, and explainability-driven validation of synthetic medical images.

Publications

The following publications are presented in APA-style format and organized chronologically.

2026

1. Das, S., & Roy Chowdhury, T. (2026). Evaluating an AI-assisted mental stress screening tool for occupational health and nursing staff. *Asian Journal of Research in Nursing and Health, 9*(1), 1754–1761. https://doi.org/10.9734/ajrnh/2026/v9i1393 .
2. Laha, N., Mazumder, P., Ghorui, L., & Roy Chowdhury, T. (2026). Intelligent PCOS prediction framework using CatBoost and AI-based clinical report generation. *International Journal of Science, Strategic Management and Technology, 2*(5), 1–9. https://doi.org/10.55041/ijsmt.v2i5.343 .
3. Sadhukhan, S., Saha, R., Das, S., Das, S., Dey, S., & Roy Chowdhury, T. (2026). The impact of artificial intelligence in modern education: A systematic framework for pedagogy, governance, and ethical co-creation. *International Journal of Science, Strategic Management and Technology, 2*(6). https://doi.org/10.55041/ijsmt.v2i6.147 .
4. Das, S., & Roy Chowdhury, T. (2026). HealthSync: A web-based smart clinical appointment and lab test scheduling system. *International Journal of Recent Development in Engineering and Technology, 15*(5), 888–893.
5. Mollah, M. B. A., & Roy Chowdhury, T. (2026). Artificial intelligence-driven face recognition in healthcare informatics: A systematic review of technologies, clinical applications, privacy challenges, and future research directions. *International Journal of Science, Strategic Management and Technology, 2*(7), 1–8. https://doi.org/10.55041/ijsmt.v2i7.007 .
6. Mollah, M. B. A., & Roy Chowdhury, T. (2026). Artificial intelligence meets healthcare informatics: A comprehensive review of object recognition for clinical decision support and smart medical systems. *International Journal Research Publication Analysis, 2*(7). DOI: 10.1555/ijrpa.3545.
7. Mandal, S., Roy, S., Das, N., Roy, A., Mandal, T., Gupta, S., & Roy Chowdhury, T. (2026). QUANTSAAS: An intelligent financial analytics and advisory platform using predictive intelligence and OCR automation. *International Journal Research Publication Analysis, 2*(7). DOI: 10.1555/ijrpa.6229.
8. Chakrabarti, S., Roy Chowdhury, T., Roy, P., & Nag, A. (2026). Advanced DeepLungCareNet: A next-generation framework for lung cancer prediction. In M. P. P. Aswathi & V. Uma (Eds.), *Intelligent Human Centered Computing* (pp. 94–105). Springer Nature. https://doi.org/10.1007/978-981-95-3671-9_9 .

2025

1. Roy Chowdhury, T., & Nag, A. (2025). Harnessing the potential of artificial intelligence to promote environmental sustainability. In *Harnessing the Potential of Artificial Intelligence to Promote Environmental Sustainability* (pp. 185–197). Emerald Publishing. https://doi.org/10.1108/978-1-83662-996-220251011 .

2024

1. Dey, S., & Roy Chowdhury, T. (2024). A comparative survey of SHAP and LIME: Explaining machine learning models for transparent AI. *International Journal of Innovative Research in Technology, 11*(6), 827–835.
2. Dey, S., & Roy Chowdhury, T. (2024). FaceCognizance: A ML based person recognition system for government surveillance in areas under possible threat. *International Journal of Creative Research Thoughts, 12*(6), b954–b960. https://doi.org/10.1729/Journal.39831 .
3. Dey, S., & Roy Chowdhury, T. (2024). The impact of serum vitamin D levels on the prevalence of musculoskeletal disorders among garment workers in Bangladesh: A cross-sectional study. *International Clinical and Medical Case Reports Journal, 3*(9), 1–31.
4. Dey, S., & Roy Chowdhury, T. (2024). Unveiling the depths: A pioneering review of deep learning models and holistic project implementations. *International Journal of Novel Research and Development, 9*(7), 604–652.

2023

1. Dey, S., Sikdar, T., & Roy Chowdhury, T. (2023). Analysis and study of cryptographic algorithm with context to cloud identity and access management. *Acta Scientific Computer Sciences, 5*(6), 28–35.
2. Dey, S., & Roy Chowdhury, T. (2023). Enhancing secure communication: Implementation and optimization of a novel algorithm for steganography with improved security and efficiency. *Acta Scientific Computer Sciences, 5*(10), 11.
3. Dey, S., & Roy Chowdhury, T. (2023). Analysis and survey of eavesdropping on cloud platform and software as a service with security. *Acta Scientific Computer Sciences, 5*(1), 130–135.
4. Roy Chowdhury, T., & Haque, S. (2023). A brief survey and analysis on satellite mobile telephone and networks. *Acta Scientific Computer Sciences, 5*(5), 26–31.
5. Dey, S., & Roy Chowdhury, T. (2023). Personalized colon cancer screening: Tailoring predictive models to individual patient profiles and genetic markers. *International Clinical and Medical Case Reports Journal, 2*(18), 1–13. https://doi.org/10.5281/zenodo.10296984 .
6. Dey, S., & Roy Chowdhury, T. (2023). Precision colon cancer screening: Leveraging data analytics and machine learning for early detection. *International Clinical and Medical Case Reports Journal, 2*(16), 1–19. https://doi.org/10.5281/zenodo.10029300 .

2022

1. Roy Chowdhury, T., Dey, P. B., et al. (2022). Analysis of cloud computing with the implementation of transmission, authenticity and broadcasting. *Gradiva Review.*
2. Roy Chowdhury, T., Prabhu, M. D., Adhikary, A., Sur, S., & Singha, P. (2022). Analysis of testing in software project environment. *International Journal of Research and Analytical Reviews, 9*(2), 391–396.
3. Jhuntu, M., & Roy Chowdhury, T. (2022). Analysis of segmentation for text components in context of scene images utilizing connected components. *Acta Scientific Computer Sciences, 4*(11), 13–17.

2020

1. Chakraborty, S., Karmakar, S., Roy Chowdhury, T., & Mahato, S. (2020). Prediction of cardiac arrest by using machine learning. *SSRN Electronic Journal.* https://doi.org/10.2139/ssrn.3649615 .
2. Rani, B., Ojha, S., Kaushal, H., Kumar, S., Roy Chowdhury, T., Karmakar, S., & Jana, B. (2020). A study on detection of brown spot disease in rice plant using machine learning technique. In *International Conference on Recent Trends in Artificial Intelligence, IoT, Smart Cities & Application (ICAISC 2020). SSRN Electronic Journal.* https://doi.org/10.2139/ssrn.3647427 .

2019

1. Mahato, S., Khatua, T., Das, A., & Roy Chowdhury, T. (2019). A brief discussion on two different cryptocurrency: Bitcoin & Ethereum. *International Journal of Computer Trends and Technology, 67*(9), 32–38. https://doi.org/10.14445/22312803/IJCTT-V67I9P106 .
2. Ghosh, A., Kumari, S., Pan, N., Kumari, S., Kumari, A., & Roy Chowdhury, T. (2019). Analysis of 5 stage pipelined operations of ARM 32/64 bit. *International Journal of Advance Research, Ideas and Innovations in Technology, 5*(1), 432–435.

2018

1. Roy Chowdhury, T. (2018). Effectiveness and classification of parallel architecture. *International Journal of Advance Research, Ideas and Innovations in Technology, 4*(3), 2455–2460.

Books and Authored Volumes

1. Roy Chowdhury, T. (2023). *A pathway to J.E.C.A.* Kindle Edition. ASIN B0C8RTG1KJ.
2. Roy Chowdhury, T. (2024). *Anomaly detection in log files: A comprehensive guide.* Lambert Academic Publishing. ISBN 978-620-7-80706-2.
3. Dey, S., & Roy Chowdhury, T. (2024). *Unveiling the Black Box: Practical Deep Learning and Explainable AI.*

Book Chapters and Scholarly Contributions

- I have contributed **12 peer-reviewed book chapters** accepted/in press with Bentham Science in edited volumes covering Artificial Intelligence, Healthcare Informatics, and Industry 5.0.
- **AI-powered OCR and Healthcare Informatics in Industry 5.0:** Book chapter contribution with Bentham Science focusing on Artificial Intelligence, optical character recognition, healthcare informatics, and Industry 5.0.
- **AI-driven Environmental Sustainability:** Book chapter contribution addressing the application of Artificial Intelligence for environmental sustainability.
- **Advanced Research Monograph:** I am developing a monograph proposal titled *Synthetic Over-Sampling for Medical Image Classification: From GANs and Diffusion Models to Clinical Deployment.*
- **MAKAUT BCA NEP Curriculum:** I am developing two textbook projects aligned with the updated MAKAUT BCA NEP curriculum:
  - Core Computational Track.
  - Advanced Application Systems.

Patents and Registered Designs

Year	Patent / Design	Details
2026	**Bio-Cipher: Privacy-Preserving Federated Learning for Genomic Data Analysis**	Indian patent application, Application No. 202631052210.
2026	**Blockchain-Secured Health Data Retrieval Device**	UK Registered Design, Design No. 6528955. Grant date: 04 June 2026.
2026	**AI-Powered Medical Device for Early Detection of Oral Cancer**	UK Registered Design, Design No. 6544078. Grant date: 17 August 2026.

Research Mentorship and Student Innovation

I actively mentor students in Artificial Intelligence, Machine Learning, Healthcare Informatics and emerging technology research.

My student research mentorship activities include:

- Biomedical waste classification using Artificial Intelligence.
- Parkinson's disease screening using Deep Learning.
- NLP-based Clinical Decision Support Systems.
- AI-based healthcare applications.
- Research-paper preparation and academic writing.
- Research publication and conference presentation.
- AI project development and implementation.
- Technical presentation preparation.

A student research team under my academic supervision received the **Best Technical Presentation Award at AMESEOM 2026**.

Conference, Reviewer and Academic Service

Role / Activity	Contribution
Reviewer – ICESAIA 2026	Reviewing international research submissions in Artificial Intelligence, Internet of Things, Healthcare Informatics, Cloud Computing and emerging technologies.
Reviewer – Analytics Global Conference 2026	Reviewing technical research contributions.
Resource Person	Parallel Architecture Workshop.
Session Chair	Multiple international conferences and technical sessions.
AMESEOM 2026	Three healthcare-AI papers accepted.
MAKAUT 2026	Participated in emerging-technology workshop and academic activities.

Workshops, Conferences and Faculty Development Activities

Date	Activity	Role
08 Oct 2018	One-day workshop on Implementation of Parallel Architecture and its Effectiveness, C.B. Seminar Hall, Techno India Chaibasa	Resource Person
10–11 Mar 2019	Information System and Technical Seminar, C.B. Seminar Hall, Techno India Chaibasa	Coordinator
15–18 Mar 2019	Three-day workshop on Big Data Analytics under TEQIP-III, conducted by NIT Warangal	Coordinator / Participant
15–18 Jul 2020	Three-day Faculty Development Program using IIT Bombay Spoken Tutorial	Coordinator / Participant
17–21 Aug 2020	FAST-2020 International Conference conducted in webinar mode	Web Expert / Session-in-Charge
05–07 Sep 2020	Webinar on COVID-19 and the pandemic situation in daily life	Web Expert / Session-in-Charge
21–27 Apr 2021	Webinar on Data Science and its Validation	Web Expert / Session-in-Charge
07–11 Apr 2022	National Conference DEEP-2022	Web Expert / Session-in-Charge / Participant

Awards, Recognition and Certifications

Award / Recognition	Details
Best Online Teacher Award (2021)	Techno Tree India, for Computer Applications subjects.
Oracle Academy Authorized Trainer	Authorized to conduct Oracle-related training in Java and Database technologies.
Best Paper Award	Recognition associated with research on secure communication and steganography.
Best Technical Presentation Award, AMESEOM 2026	Achieved by a student research team under my academic supervision.
First Position in Tabla Playing	Bengal Music College, Sodepur (2005).
5-Star Educator Status	TeacherOn / UrbanPro.
DEEP-2022 Best Paper Award	Recognition associated with the National Conference DEEP-2022.

Teaching and Training Profile

My teaching experience covers Computer Science and Engineering, programming, Artificial Intelligence, Machine Learning, networking, databases, web technologies, software engineering, computer architecture, and emerging technologies.

Areas of teaching and training include:

- C and C++ Programming
- Java Programming
- Python Programming
- Data Structures and Algorithms
- Database Management Systems
- Computer Networks
- Computer Architecture
- Web Technologies
- Software Engineering
- Machine Learning
- Artificial Intelligence
- Deep Learning
- Cloud Computing
- ERP Systems
- Emerging Technologies

I have more than a decade of teaching and academic experience and have also conducted professional and online teaching activities through platforms including TeacherOn and UrbanPro.

Curriculum Development and Academic Content Creation

- Development of academic notes, teaching materials and technical resources.
- Curriculum development for Computer Science and emerging technology subjects.
- Development of educational materials aligned with university curricula.
- Current development of two textbook projects for the MAKAUT BCA NEP curriculum.
- Preparation of practical programming and technology-oriented learning materials.
- Development of AI and healthcare research resources.
- Research-paper mentoring and technical academic writing.

Professional Skills

Academic and Research Skills	**Technical and Professional Skills**	
Academic Administration	AI/ML System Development	
Departmental Coordination	Healthcare Technology Prototyping	
Research Supervision	Web Administration	
Student Project Mentoring	ERP Management	
Technical Writing	Institutional Technology Management	
Research-Paper Development	Conference Organization	
Curriculum Design	Session Management	
Academic Content Development	Technical Reviewing	
Student Innovation	Research Presentation	

Professional and Academic Engagements

- Assistant Professor and academic researcher in Computer Science and Engineering.
- PhD researcher in Health Informatics at NIT Silchar.
- Oracle Academy Authorized Trainer.
- Reviewer for international conferences.
- Session Chair for international technical conferences.
- Resource Person for technical workshops.

- Academic Advisor and departmental coordinator.
- Research mentor for undergraduate AI and healthcare projects.
- Academic content developer and curriculum contributor.
- Web and ERP systems administrator.
- Student research publication and conference mentor.

Research Impact and Academic Contributions

My academic research has progressively evolved from Computer Architecture, Parallel Processing, Software Testing, Cloud Computing and Security toward Artificial Intelligence, Machine Learning, Healthcare Informatics, Medical Imaging, Explainable AI and Federated Learning.

My recent research portfolio emphasizes:

- AI-driven healthcare prediction.
- Privacy-preserving healthcare intelligence.
- Explainable and trustworthy AI.
- Medical-image analysis.
- Cancer prediction and screening.
- Genomic data analytics.
- Federated Learning.
- Synthetic medical-data generation.
- Clinical decision-support systems.
- Generative AI for healthcare.
- AI-based educational and institutional systems.

Personal Information

Particular	**Details**
Date of Birth	29 March 1990
Gender	Male
Marital Status	Married
Father's Name	Mr. Jayanta Roy Chowdhury
Mother's Name	Mrs. Mithu Roy Chowdhury
Spouse	Mrs. Rhijurekha Roy Chowdhury
Son	Trishaan Roy Chowdhury
Address	Flat No. 3B, Renuka Palace, Vivekananda Pally, Taki Road, Barasat, Kolkata–700124, West Bengal, India
Mobile	+91 8902503176; +91 7003585265
Email	tathagatamtech13@gmail.com; tathagata.r@tecb.edu.in

Hobbies and Interests

Playing cricket and volleyball; watching movies; singing; playing tabla and music-related activities.

Declaration

I hereby declare that the information presented in this curriculum vitae is a true representation of my academic qualifications, professional experience, teaching activities, research contributions, publications, patents, academic service, awards, and scholarly activities to the best of my knowledge.

Place: Kolkata **Date:** 27 August 2026

TATHAGATA ROY CHOWDHURY
Kolkata,West Bengal, India